

# Project Demo Planning

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Date          | 15 July 2026                                                   |
| Project Name  | Rising Waters: A Machine Learning Approach to Flood Prediction |
| Maximum Marks | 1 Mark                                                         |

This document outlines the plan for demonstrating Rising Waters, including the flow of the demo, key features to highlight, and timing. As a solo project, all sections are presented by a single presenter.

| S<br>·<br>N<br>o | Demo Section                     | Description                                                                      | Duration (mins) | Responsible Member |
|------------------|----------------------------------|----------------------------------------------------------------------------------|-----------------|--------------------|
| 1                | Introduction                     | Frame the problem: lack of affordable, hyperlocal flood early-warning tools      | 2               | Nanda Gunasri      |
| 2                | Solution Overview                | Walk through the Rising Waters architecture and the rainfall-to-risk ML pipeline | 3               | Nanda Gunasri      |
| 3                | Live Demo — Safe Scenario        | Enter low-risk rainfall values and show the “Low Flood Likelihood” result        | 3               | Nanda Gunasri      |
| 4                | Live Demo — Flood-Risk Scenario  | Enter high-risk rainfall values and show the flood-risk result                   | 3               | Nanda Gunasri      |
| 5                | Validation & Error Handling Demo | Submit invalid input to show graceful error handling without a crash             | 2               | Nanda Gunasri      |
| 6                | Q&A / Roadmap Discussion         | Discuss the Phase 2 roadmap: live APIs, alerts, GIS mapping                      | 2               | Nanda Gunasri      |

## Demo Flow Summary

| Step | Activity                         | Notes                                                                                                      |
|------|----------------------------------|------------------------------------------------------------------------------------------------------------|
| 1    | Introduction & Problem Statement | Explain why smaller flood-prone communities lack affordable early-warning tools                            |
| 2    | Solution Overview                | High-level walkthrough of the ML pipeline (rainfall data → XGBoost → risk %) and system architecture       |
| 3    | Live Feature Demonstration       | Run a safe prediction and a flood-risk prediction live on the deployed app, plus one invalid-input example |
| 4    | Q&A Session                      | Open floor for questions on model choice, validation approach, and the future roadmap                      |